An investigation on the relationship between slit distance and fringe spacing

Jimmy Zeng

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Contents

1 Introduction and Background

Wave Phenomena states that wave could interfere when they overlap. When two waves are in opposite phase, it is said to interfere destructively and cancel out. However, if they are the same phase and interferes, it is said to be constructive, reaching a greater amplitude.

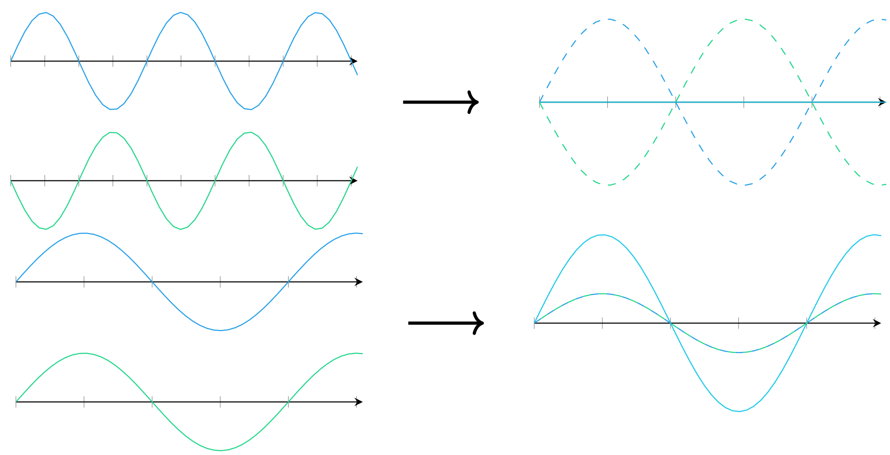

Figure 1: Illustration of wave interference

The debate of whether light is a particle or a wave was challenged by Thomas Young's famous Double Slit Experiment in Young's Course of Lectures on Natural Philosophy and the Mechanical Arts of 1807, where a coherent light source shinning through two tiny slits projected an interference pattern with alternation between light and dark fringes rather than two individual bright fringes in a dark background. Illustrated In Figure 2

Figure 2: A Double Slit System

The formula proposed by Thomas Young was

$$s = \frac{\lambda D}{d} \tag{1}$$

In which "S" is the distance between two adjacent bright fringes, λ is the wavelength of the light, D is the distance between the screen and slits in meters and d is the distance between slits. The mathematical geometric derivation is thus justified: Consider a point a on the screen of a double slit system at which two light rays touch from their respective slits. This is illustrated in figure 3

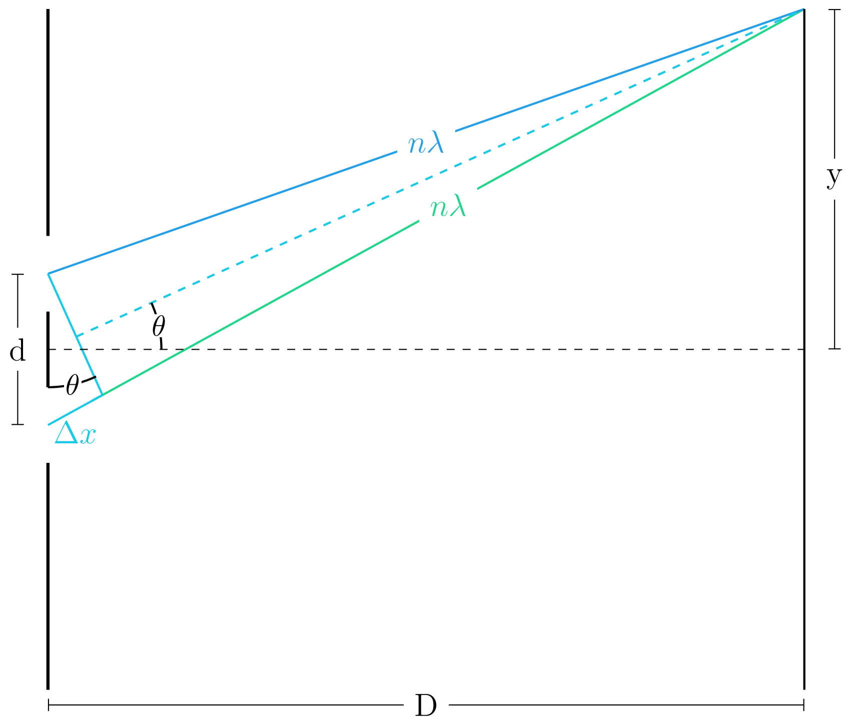

Figure 3: Zoomed in view of double slit system

We observe that the θ that is related by the equation

$$\sin(\theta) = \frac{\Delta x}{d} \tag{2}$$

is equal to the θ represented by

$$\tan(\theta) = \frac{y}{D} \tag{3}$$

When the distance of both light rays' traveled paths is a multiple of the wavelength of the light, they engage in constructive interference. We observe the fact that the path difference (Δx) for constructive waves is

$$n\lambda \tag{4}$$

where λ is the wavelength of the light and n is an integer representing the interested bright fringe. Hence we derive the equation

$$\sin(\theta) = \frac{\Delta x}{d} \tag{5}$$

$$\sin(\theta) = \frac{n\lambda}{d} \tag{6}$$

$$\theta = \sin^{-1}\left(\frac{n\lambda}{d}\right) \tag{7}$$

and since we know that the wavelength λ of light is orders of magnitudes larger than slit spacing d, and hence the value in sin is close to zero, we can use the sin angle approximation $\lim_{\theta \to 0} \sin(\theta) = \theta$ to simplify to

$$\theta = \frac{n\lambda}{d} subsubsection \tag{8}$$

relating θ to d and λ. This angle approximation for small values of θ also apply to $\tan(\theta)$ in the other representation of θ

$$\tan(\theta) = \frac{y}{D} \tag{9}$$

and since we know that this is the same θ we can simplify to

$$\theta = \frac{y}{D} \tag{10}$$

and putting them together

$$\frac{y}{D} = \theta \tag{11}$$

$$\frac{n\lambda}{d} = \theta \tag{12}$$

$$\frac{y}{D} = \frac{n\lambda}{d} \tag{13}$$

$$y = n\frac{D\lambda}{d} \tag{14}$$

$$\tag{15}$$

to find the distance between fringes we need to find the distance of a fringe to the central maxima and distance of a consecutive fringe to the central maxima and find the the difference between their distances.

We have defined the distance from the central maximum to

- the nth fringe as $y = n\frac{D\lambda}{d}$
- the nth + 1 fringe as $y = (n+1))\,\frac{D\lambda}{d}$

finding the difference $\Delta y = s$ gives us the spacing between adjacent fringes

$$\Delta y = s = (n+1)\,\frac{D\lambda}{d} - n\frac{D\lambda}{d} \tag{16}$$

$$s = (n+1-n)\,\frac{D\lambda}{d} \tag{17}$$

$$s = \frac{D\lambda}{d} \tag{18}$$

and now we arrive at our destination.

2 Methodology

2.1 Research Hypothesis and Method variables

The independent variable is the distance D between the slit and the screen. There will be two experiments carried out with different setups, namely, the ”macro” and ”micro” experiments. This is to satisfy the trial quantity to increase validity and the nature of each experimental procedure, which is explained in the ensuing sections.

- Micro will solely focus on distances within a meter and will include (in cm) 20, 40, 60, 80, and 100.
- Macro will be done from various distances ranging from 3 to 8 meters and will include (in m) 3, 5, 6, and 8.
  - Each of these distances will have micro steps that include (in meters) +0, +50, and +100 which will extend our trial amounts
  - For example, for 5 meters we will have (in m) 5.00, 5.50, and 6.00.

What is the relationship with the distance between screen and slit (from the range of 20cm to 8 m) and the spacing between adjacent bright fringes (in mm) in a double slit system.

The measured dependent variable is the distance s between two adjacent bright fringes. Illustrated by the following diagram.

And hence our guiding question is

And we hypothesize that the relationship would be a linear relationship with the distance between screen and slit and the spacing between adjacent bright fringes. This is because our mathematically derived equation $s = D\frac{\lambda}{d}$ suggests a directly proportional relationship between s and D.

2.2 Variable Control and Measurement

The distance between slit and screen will be modified two means.

The long range consists of the aforementioned 3m, 5m, 6m, and 8m. It will be achieved with the placement of three tables (of which the recording instrument will hang on), the last 8m is the distance from the origin directly to the wall. They will be relative to an arbitrary "origin" point in the classroom marked with a masking tape.

The short range will represent the aforementioned "micro steps" (0cm, 50cm, and 100cm) and will be achieved by placing the slit (and light source) on a momentum test cart (this is because they move back and forth very smoothly on their designated tracks).

This range of data will provide us justification for not only the short range, but also long range to show the validity ofthe double slit equation over a range of distances.

The distance between adjacent fringes will be measured by:

1. Having the interference pattern projected on a graphing paper with a known grid size
2. Taking a high resolution photo of the graphing paper
3. Perspective correction of the image
4. Data measurement by software means.

(a) Finding the amount of pixels per cm across different parts of the image

(b) Set the average as the scale for measurements

(c) Test accuracy by measuring the 1mm grid on the paper with set scale

(d) Measure the distance between left and right edges of a fringe (the immediate intermediate between bright and dark fringes)

(e) Repeat last step for each maxima

(f) Repeat steps 1-5 for each image of each distance between slit and screen

The following control variables are considered:

- Using the same laser-slit instrument
- Same method for setting up each table
- same method for measuring fringe distance

2.3 Setup Description and Experimental Procedure

The setup will consists of the table components and the slit system. The slit system is a laser-slit system mounted on a two axis translation stage. This allows for changing the distance without moving away the table but rather by moving sideways and also allows for more microadjustments for each table. This is built with:

- Two axis translation stage
  - 3 Momentum test tracks (Ours in particular had centimeter markings on the side and were perfectly cut off from 0cm)
  - 3 Momentum test carts
- Laser-slit instrument
  - Make sure to measure the offset distance between the slit to the edge at the bottom of the instrument that you will use as the instrument's "origin point"

- 4 stacks of books

The table will have a graphing/grid paper on a clipboard that is then hung on a part of the table with a setup displacement (described later in detail) Required components for this build include:

- Masking tape (or one you can write on)
- Table (duh)
- Clipboard
- Graphing/Grid paper (of which the grid is of uniform and known length)

2.3.1 Setup Procecdure

Additional equipment required for accurate setup is as listed:

1. Set squares (or likewise)
2. Retort stand
3. Meter stick
4. Masking tape
5. Markers
6. Rulers
7. Measuring tape ($\tilde{8}$m)

1. Taping the end of the measuring tape onto the floor with masking tape
2. Extend to the end of the wall
3. Stack the books on the floor like shown
4. Place the two cart tracks atop of each pair of books perpendicular to the direction of the ray

5. Align the forward facing (towards the screen) cart track with a ruler (or any straight object) facing perpendicular to the ground to ensure the outermost edge is aligned with the origin.
6. Place carts on each track
7. Place a cart track atop of both carts (parallel to the direction of the ray)
8. Place a cart on the cart track
9. Place the double slit laser instrument on the cart
10. Tape the double slit laser instrument onto both sides of the cart
11. Choose a designated "origin point" (that acts as a reference point for its distance) and discover its' horizontal offset distance from the slit
12. Place a meter stick perpendicular to the measuring tape.
    - Make sure the edge facing away from the laser is aligned with the desired meter mark. This makes it easier to more accurately place the tables.
    - ensure that the stick is as perpendicular to the measuring tape as possible. This ensures both sides of the table is accurate to the desired displacement. This can be achieved with a set square.
13. Place the table near the interested displacement
14. Place a retort stand upside down as illustrated on one end of the table
15. Using a ruler, extend the length of measurement by placing it on the side of the rod away from the laser.
16. Move that end of the table until it aligns with the desired displacement
    - Ensure the side of the ruler facing the laser touches the side of the meter stick facing away from the laser.
    - Ensure that the ruler has no gaps and is perfectly parallel to the iron rod.

- Since the surface that is to be projected on is the surface opposite from the laser, placing the ruler on the other side ensures the side of the ruler facing the laser represents the distance of the surface being project on. and by ensuring the surface of the meter stick away from the laser (representing the desired distance to be projected on) is touching the ruler (representing the distance of the screen to be projected on), we ensure that the table is perfectly positioned on the desired distance.

17. Move the retort stand to the other side of the table and re-enact the alignment ritual

18. Repeat for each table and stand feeling proud of collecting all the infinity stones

2.3.2 Experimental Procedure

Recall the two changing systems that comprise the control of the independent variables After the setup is complete begin the following experimental procedure:

1. Ensure no pedestrians are abode the path of the laser

2. Laser, go!

3. Adjust the momentum test cart such that the [set] origin point of the laser-slit system aligns with your desired distance from the origin point of the momentum test track

4.

Since the fringe spacing is within the milimeter range and it is beyond my human ability to hold the clipboard still within milimeter stillness while using the other hand to hold a pencil and mark accurately on the paper live. I decided to take a high resolution photo of the projection on a graphed piece of paper as a photo is most versatile and captures exactly as seen. The graphed piece of paper is printed with milimeter accuracy markings. As a result I will measure distance between each adjacent fringe (in milimeters) with software solutions and excuse my skill issue.